

Prove one correction loop before scaling the system.

The first two passes create a shared correction contract, replayable evidence, and a narrow shadow loop. The goal is to reduce uncertainty without using the machine floor as an avoidable test harness.

DAYS 0-15	Select the first correction class Choose a problem with customer value, measurable outcomes, sufficient evidence, bounded consequence, and platform leverage.	Exit: named owner, target state, success criteria, hold conditions.
DAYS 16-30	Build the correction contract Define intent, state, coordinate frames, data sources, cause hypotheses, action limits, authority, validation, and reuse scope.	Exit: replayable baseline and signed correction object.
DAYS 31-45	Build the evidence harness Create deterministic replay, simulation inputs, state visualization, experiment tracking, and a representative regression set.	Exit: repeatable test path before machine action.
DAYS 46-60	Close the shadow loop Reconstruct state, rank causes, propose bounded corrections, compare with expert action, and plan post-action proof.	Exit: disagreement log, rollback, before/after evidence plan.

TECHNICAL CHECKPOINTS

Checkpoint	Question	Evidence
Intent fidelity	Can the system recover what the operation is trying to accomplish?	Trace from CAD/CAM/G-code/context to explicit intent.
Spatial truth	Are machine, tool, part, fixture, sensor, and metrology frames aligned?	Registration error, lineage, calibration state, and frame owner.
Model/physics split	What is learned, simulated, constrained, or deterministic?	Ablations, residual analysis, and known failure modes.
Authority	What can observe, recommend, execute, or stop?	Release rules, fallback, override, incident, and rollback.

Stress the boundary, then install the compounding system.

DAYS 61–75	Stress the operating envelope Test warm-up, environment, tool and material variation, fixture and coordinate changes, program style, missing data, sensor degradation, and rollback.	Exit: failure taxonomy and known limits.
DAYS 76–90	Qualify the release posture Compare shadow recommendations with expert decisions, establish post-correction proof, and decide whether the class remains shadow, earns bounded release, or holds.	Exit: explicit release and incident rules.
DAYS 91–105	Make architecture boundaries durable Install data and frame contracts, model registry, simulation interface, evaluation, observability, lineage, retention, and rollback.	Exit: first proven platform contracts.
DAYS 106–120	Build the founding AI operating cadence Clarify ownership, hiring priorities, experiment review, correction review, architecture decisions, product evidence, and correction-class portfolio choices.	Exit: team design and next two correction classes.

INITIAL SCORECARD

Dimension	Evidence to track	Decision supported
Correction quality	Post-action metrology, repeatability, residuals, recurrence, secondary effects.	Did the action improve the intended result?
Evidence velocity	Time from deviation capture to qualified disposition; experiment reproducibility.	Where should infrastructure remove delay?
Containment	Unsafe proposals blocked, ambiguity detected, fallback, rollback, audit completeness.	May authority expand?
Portability	Reuse by machine, tool, material, part family, operation, and environment.	What becomes a platform capability?
Operator trust	Overrides, accepted recommendations, explanations, owner clarity, training burden.	Is adoption improving with reliability?
Company leverage	Roadmap decisions, hiring clarity, customer proof, demo credibility, architecture reuse.	What should 4MP build next?

Day 120 should produce a proven correction class, durable platform contracts, a visible evidence cadence, and focused hiring decisions—not a broad AI roadmap detached from machine proof.